

Intoduction To Programming Using C++ Level 2

Register Variable



Introduction To Programming Using C++

Level 2

Register Variable

Definition

- ◆ A Register Variable is a Variable that was traditionally requested to be stored in a CPU Register instead of regular RAM, with the goal of providing faster access during program execution.
- ◆ Register Variables were commonly associated with Variables that are accessed very frequently, such as Loop Counters.

Explanation

- ◆ In a typical program, Variables are stored in memory managed by the system, and the Compiler decides how they are actually placed during execution.
- ◆ The `register` Keyword was historically used to suggest to the Compiler that a Variable should be kept in a CPU Register because Registers are very fast.
- ◆ However, modern C++ Compilers perform their own optimization and usually decide automatically which Variables should be placed in Registers.
- ◆ Therefore, the `register` Keyword is obsolete in modern C++ and should not be used.

Procedures

1. Disk / SSD

- Used for persistent storage.
- Much slower than RAM and CPU memory.

2. RAM

- Used as the main working memory of a running program.
- Faster than persistent storage but slower than CPU Registers and Cache.

3. CPU Cache

- Very fast memory located close to or inside the CPU.
- Stores frequently accessed data to reduce access time.

4. CPU Registers

- Very small and extremely fast storage locations inside the CPU.
- Used by the processor to hold values that are actively being processed.

💡 Example

```
int main()
{
    register int Counter = 0;

    for (Counter = 0; Counter < 10; Counter++)
    {
        cout << Counter << endl;
    }

    return 0;
}
```

➡ Result

0
1
2
3
4
5
6
7
8
9

⊘ Warning

- ◆ In modern C++, ``register`` is deprecated and should not be used in new code.
- ◆ The important concept to understand is the role of CPU Registers and Compiler Optimization, rather than using the ``register`` Keyword itself.



Comparison

▨ Register Variable

- ✓ Historically used to request CPU Register storage.
- ✓ Intended for frequently accessed Variables.
- ✓ Could potentially reduce access time.
- ✓ The Compiler was not required to follow the request.

▨ Modern C++

- ✓ The Compiler automatically performs Register Allocation.
- ✓ The Programmer normally does not need to specify `register`.
- ✓ Modern optimization is generally more effective than manually requesting Register storage.

✦✦ Summary

- ✓ CPU Registers are very small and extremely fast storage locations inside the CPU.
- ✓ The ``register`` Keyword was historically used to suggest storing a Variable in a CPU Register.
- ✓ It was commonly associated with frequently accessed Variables such as Loop Counters.
- ✓ ``register`` never guaranteed Register storage.
- ✓ Modern Compilers automatically decide how to optimize Variable storage.
- ✓ ``register`` is deprecated in modern C++ and should not be used in new code.

Thank you 😊